

3.18 BUILDING CONSTRUCTION (446)

Building Construction Paper

SECTION A **Marks**

Answer all the questions in this section in the spaces provided.

- [illegible]

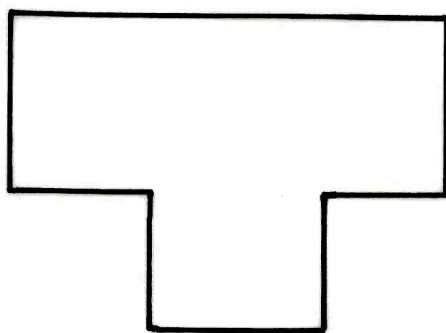


Figure 3.3

- Make sketches to show the positions of the profile boards for setting out the building.



□□ State **Two** functions of each of the following materials in concrete □

□ **a** water □ □ □ □ □ □ □ **10 marks** □

□ **b** coarse aggregate □ □ □ □ □ □ □ **10 marks** □

□□ **a** Outline **four** functions of foundations in buildings □ □ **4 marks** □

□ **b** State **four** factors that are considered when choosing a type of a foundation □ **4 marks** □

□□ State **four** ways of controlling termite on site □ **4 marks** □

□□ **a** Give **four** reasons for bonding a wall □ □ **4 marks** □

□ **b** Sketch an elevation of a wall built in English bond to show four courses high and four bricks in length with a stopped end □ **4 marks** □

□□ **a** State **four** factors which influence the selection of a roof of a building. (2 marks)

□ **b** Sketch and label a section through a lean to roof □ □ **4 marks** □

□□□ Figure **1** shows the views of a shaped block in **1st** angle projection □

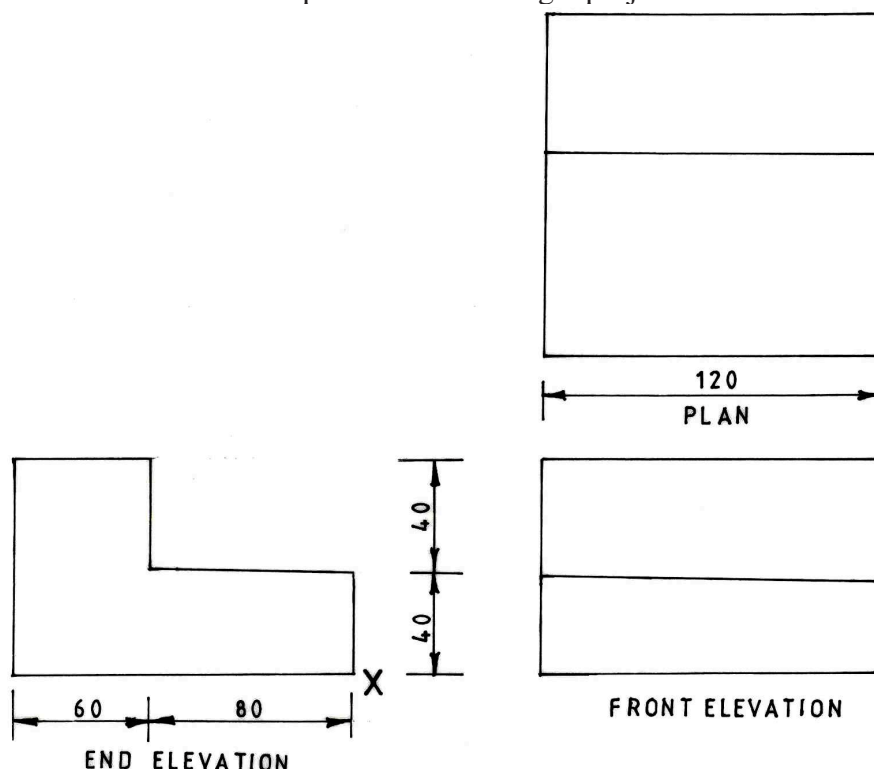


Figure **1**

□ Using the scale of **1:100** draw the block in isometric projection with **X** as the lowest point □ **10 marks** □



SECTION B (10 marks)

Answer Question 1 on the A paper provided and any other three questions from this section in the spaces provided after Question 1. Candidates are advised not to spend more than 10 minutes on a question.

Figure 1 shows a pictorial view of a shaped block.

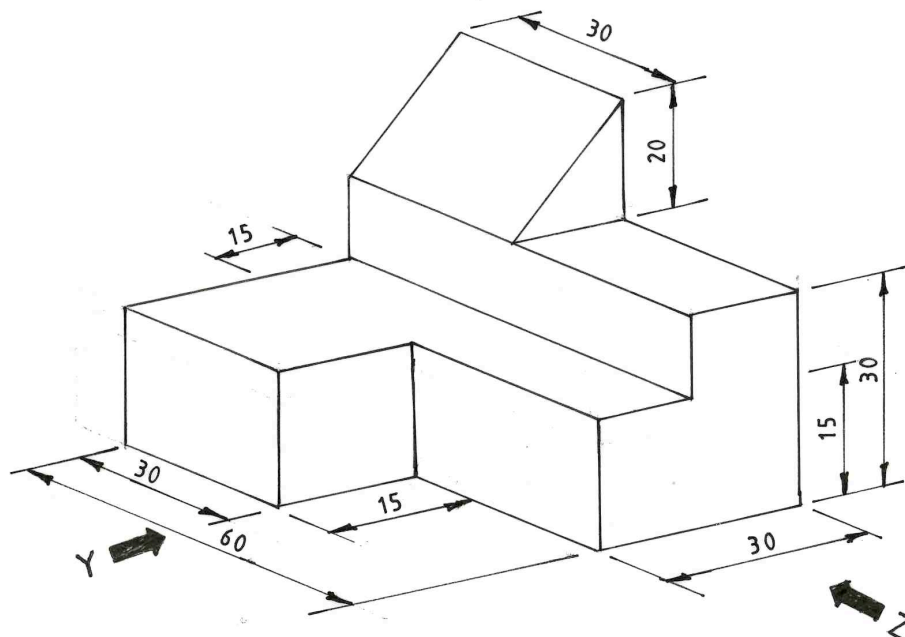


Figure 1

Draw full size, in first angle projection the following views of the block:

- ☐ ☐ (i) front elevation in the direction of arrow Y
- ☐ ☐ (ii) end elevation in the direction of arrow Z
- ☐ ☐ (iii) plan

(10 marks)

(a) Explain each of the following purposes of finishes in buildings:

- ☐ ☐ (i) protection of surfaces
- ☐ ☐ (ii) hygiene
- ☐ ☐ (iii) insulation

(6 marks)

(b) Outline the procedure of preparing a newly plastered surface to receive emulsion paint.

(4 marks)

(a) Outline the procedure of laying a raised concrete floor slab. (8 marks)

(b) Sketch and label a cross-section of a timber stair to show the joint details of two steps.

(6 marks)

(c) Using a vertical sectional sketch show four positions where damp proofing is provided in a building.

(4 marks)

(d) Sketch and label a vertical section of a dependent scaffold.

(4 marks)

(e) Sketch and label a cross-section of a direct domestic hot water supply system, up to the first floor by incorporating an immersion heater. (15 marks)